

Abinand K Vinod

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EDUCATION

College of Engineering Munnar, Munnar

Aug 2018 – May 2022

Bachelor of Technology in Computer Science

Luminar Technohub, Ernakulam

May 2023 – Jan 2024

Data Science, Python, and Machine Learning

EXPERIENCE

AI/ML Engineer Consultant

Feb 2025 – Jan 2026

Awans Smart Solutions, Thiruvananthapuram, Kerala

- Built a real-time multi-camera semantic segmentation system to automate driving test pass/fail evaluation for Kerala MVD.
- Designed and deployed low-latency video inference pipelines using PyTorch and OpenCV for live processing.

AI/ML Research Engineer Trainee

Apr 2024 – Dec 2024

Zoho Research and Development Labs, Kottarakkara, India

- Researched and implemented approximate matrix multiplication (MADNESS) techniques in C++ for performance-critical workloads.
- Benchmarked and optimized ML kernels for latency reduction and memory efficiency.

Data Science Intern

Jan 2024 – Mar 2024

Luminar Technohub, Kakkanad, India

- Developed end-to-end machine learning pipelines on real-world datasets from preprocessing to evaluation.
- Built and validated supervised ML and deep learning models using industry-standard metrics.

PROJECTS

MLOps Project – Vehicle Insurance Data Pipeline | *Python, MongoDB, AWS, Docker, GitHub Actions*

- Built an end-to-end MLOps pipeline covering data ingestion from MongoDB, validation, transformation, model training, evaluation, and deployment using modular Python components.
- Implemented CI/CD automation with Docker and GitHub Actions; deployed models on AWS EC2 with S3-based model registry and ECR-backed container delivery.

COVID-19 Chest X-Ray Classification | *Python, PyTorch, EfficientNet*

- Trained an EfficientNet-based transfer learning model to classify chest X-rays into COVID-19, Pneumonia, Lung Opacity, and Normal classes.
- Achieved 92.3% validation accuracy using fine-tuning and learning rate scheduling.

Multi-PDF Chatbot | *Python, Streamlit, LangChain, Gemini Pro, FAISS*

- Built a RAG-based chatbot capable of answering queries across multiple uploaded PDFs with conversational memory.
- Implemented FAISS-based vector retrieval for fast and accurate semantic search.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Neural Networks

Computer Vision: OpenCV, CNN, LSTM

GenAI / LLM: LangChain, RAG, LangGraph, HuggingFace

MLOps: Model Deployment, Experiment Tracking, CI/CD, Docker

Tools: Git, Streamlit, MCP